



Fate and Transport of Polar Organic Micropollutants in Soil–Plant Systems Quantified by Mass Balance using Laboratory-Scale Lysimeters

Pascal Hasselder · Han Mao ·
Aki Sebastian Ruhl

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Abstract Irrigation with reclaimed water can mitigate water scarcity but may also introduce polar organic micropollutants (OMPs) into agricultural systems. As these compounds are frequently detected in treated wastewater and aquatic environments, this study investigated their environmental fate and transport in the vadose zone, with emphasis on processes in an unsaturated soil–plant system. Spinach was cultivated in laboratory-scale lysimeters irrigated with spiked water, followed by leaching with artificial rainwater. A mass-balance approach was used to quantify the relative contributions of plant uptake, soil retention, leaching, and transformation. Across the fourteen investigated OMPs, soil retention was the dominant process for three substances, leaching for five, and transformation for three, while the remaining compounds showed mixed behavior dominated by leaching and transformation. Trifluoromethanesulfonate exhibited distinctly higher plant uptake than the other OMPs; however, plant uptake

contributed only marginally to overall mass balances. These results provide process-based insight into the fate of polar OMPs in soil–plant systems and highlight the potential for rapid leaching of polar OMPs to groundwater. The findings contribute to improved assessment of contaminant transport and support risk management of agricultural water reuse.

Keywords Polar organic micropollutants · Water reuse · Vadose zone · Groundwater · Leaching · Transformation

1 Introduction

Increasing water stress is challenging different sectors of human activity worldwide, and the use of reclaimed treated wastewater for agricultural irrigation has been adopted as a mitigation strategy (Christou et al., 2024; Van Vliet et al., 2021). Besides its benefits, this practice may introduce environmental risks, including incomplete removal of organic micropollutants (OMPs) during conventional wastewater treatment and subsequent release into the environment (Christou et al., 2024; Falås et al., 2016; Helmecke et al., 2020).

Persistent and mobile OMPs have received increasing attention as a particularly concerning group of OMPs. While the European Regulation concerning the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH; EC 1907/2006)

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P. Hasselder (✉) · A. S. Ruhl
Section of Water Treatment (II 3.3), German Environment Agency, Schichauweg 58, 12307 Berlin, Germany
e-mail: pascal.hasselder@uba.de

P. Hasselder · H. Mao · A. S. Ruhl
Chair of Water Treatment, Technische Universität Berlin, Straße Des 17. Juni 135, 10623 Berlin, Germany

provides criteria for persistence based on degradation half-lives in soil, sediment, and water, it does not include a mobility criterion. Since 2023, persistent, mobile and toxic (PMT) and very persistent, very mobile (vPvM) substances have been recognized as distinct hazardous substance classes under the European Classification, Labelling and Packaging (CLP) Regulation. The P and vP persistence criteria under CLP are identical to those under REACH, with degradation half-lives of more than 120 and 180 d in soil defining P and vP, respectively. Under CLP, the mobility criteria are fulfilled when the logarithm of the organic carbon–water partition coefficient ($\log K_{oc}$) is lower than 3 and 2 for M and vM, respectively. For ionizable substances, the lowest $\log K_{oc}$ value between pH 4 and 9 is applied. The pH dependent octanol–water distribution coefficient ($\log D$) has been discussed as surrogate for mobility in soil and water as it indicates hydrophilicity and polarity. When experimental data of K_{oc} or $\log D$ is missing, modeled $\log D$ values are commonly applied to estimate mobility of OMPs (Berger et al., 2018; Hale et al., 2020). Typically, substances with $\log D$ below 2 are considered polar. Polar substances are frequently detected in aquatic environments and may persist even in advanced wastewater treatment systems as well as during natural attenuation processes (Muschket et al., 2024; Neuwald et al., 2023; Zee-shan & Ruhl, 2023). Many chemicals that are used in households (e.g., pharmaceuticals, personal care products, cleaning agents) and industrial applications (e.g., processing agents and additives), exhibit physicochemical properties associated with persistence and mobility (Arp & Hale, 2019). Consequently, the exposure of agricultural soils to polar OMPs following irrigation with reclaimed water or surface water that is impacted by treated wastewater is expected.

In irrigated soils, OMPs might undergo transformation, sorption, leaching to groundwater, or uptake by plants (Ben Mordechay et al., 2021; Biel-Maeso et al., 2018; Carter et al., 2014; Conn et al., 2009; Denora et al., 2024; Sallwey et al., 2020). The polarity and ionic speciation of OMPs influence their interactions with soil constituents and might lead to high mobility and leaching potential, posing risks to groundwater quality, which is a resource for drinking water supply in many regions. Conversely, retention of persistent and polar OMPs in soil might increase their residence time, potentially resulting in

delayed transport, prolonged exposure, or accumulation when inputs exceed removal processes. Although polar OMPs generally exhibit lower partitioning into soil organic matter than hydrophobic compounds, their mobility can be moderated by specific interactions with soil mineral and organic surfaces, including electrostatic interactions, ion exchange, and surface complexation (Strawn, 2021; Thalmann et al., 2025a, b). Retention may enhance transformation by increased residence times before OMPs reach groundwater. Biofilms can provide additional retention sites through ionic and other interactions within the extracellular polymeric matrix, while simultaneously acting as reactive zones for OMP transformation (Flemming & Wingender, 2010; Jou-Claus et al., 2024). Generally, biotransformation is considered a process governing transformation of OMPs in soil (Greskowiak et al., 2005; Maeng et al., 2011).

OMP plant uptake is a multifaceted process influenced by environmental conditions of solute and soil matrix, as well as physicochemical properties of OMPs. While root uptake of non-polar and moderately polar OMPs is generally considered to negatively correlate with adsorption to soil organic matter, uptake of polar and ionized compounds is less understood (Miller et al., 2016). Several studies have demonstrated the uptake of polar OMPs by leafy vegetables and other crops (Manasfi et al., 2021; Riemen-schneider et al., 2016; Seelig et al., 2025). However, polarity and ionic speciation were reported to insufficiently predict plant uptake or translocation among a set of polar OMPs (Seelig et al., 2025).

Previously, concerns regarding polar OMPs in the context of irrigation with reclaimed water have been raised (Helmecke et al., 2020). However, their fate in soil–plant systems remains insufficiently understood. Integrated assessments of multiple fate pathways of polar OMPs are still scarce, limiting robust environmental risk assessment.

Accurate mass balances enable disentangling the pathways of sorption, leaching, plant uptake, and transformation, where transformation is defined as the mass fraction not recovered in the quantified compartments, potentially reflecting transformation processes or unaccounted retention. Field studies with complex boundary conditions make accurate mass balancing difficult, and large-scale lysimeter experiments are resource-intensive, both limiting systematic investigation of fate pathways. Laboratory-scale lysimeters

provide a controlled approach to investigate contaminant fate and enable pathway-oriented assessments.

This study investigates the fate of selected twelve polar and two non-polar OMPs in a soil–plant system using controlled laboratory-scale lysimeters and arable soil. Spinach was chosen as crop due its common use as leafy vegetable, its short cultivation time and because leafy vegetables tend to exhibit higher OMP uptake compared to other crops (Christou et al., 2019). Spinach cultivation was carried out under controlled conditions and irrigation with water containing the selected potentially OMPs. Subsequently, artificial rainwater was applied to simulate leaching during heavy rainfall. By quantifying OMP masses in leachates, at different soil depths, and in plant material, pathways governing the environmental fate are identified.

2 Materials and Methods

2.1 Experiment Setup

The experimental setup comprised three groups, each consisting of four vegetated and two non-vegetated (reference) laboratory-scale lysimeters, which were

subjected to the same experimental phases (Fig. 1). Lysimeters were prepared from polyvinyl chloride tubes (inner diameter 10.5 cm, length 22.5 cm), of which 15 cm were filled with soil. The soil was collected from the upper 30 cm of an agricultural site in southern Brandenburg, Germany (51°36'24.2"N 13°18'07.1"E) that was not irrigated or exposed to the OMPs before. It was classified as coarse sand (95% sand, 4% silt, 1% clay), with a pH of 5.3 (0.1 mol/L CaCl_2), a total organic carbon content of 1.2% (g/g), and a grain density of 2.65 g/cm³. Lysimeters were packed to a dry bulk density of 1.5 g/cm³, corresponding to an estimated porosity of 43.8% (v/v). A 2 cm thick drainage layer of gravel was installed at the lysimeter's bottom to ensure free percolation through perforated end caps.

Fourteen OMPs were selected to represent substances from pharmaceutical, household, and industrial applications that have been repeatedly detected in treated wastewater at levels ranging from sub- $\mu\text{g/L}$ to tens of $\mu\text{g/L}$ (Finckh et al., 2022; Muschket et al., 2024; Neuwald et al., 2023; Schulze et al., 2019). Based on literature data and screening of substances registered under the European REACH regulation, most of these OMPs are considered potential persistent and mobile substances (Berger et al., 2018).

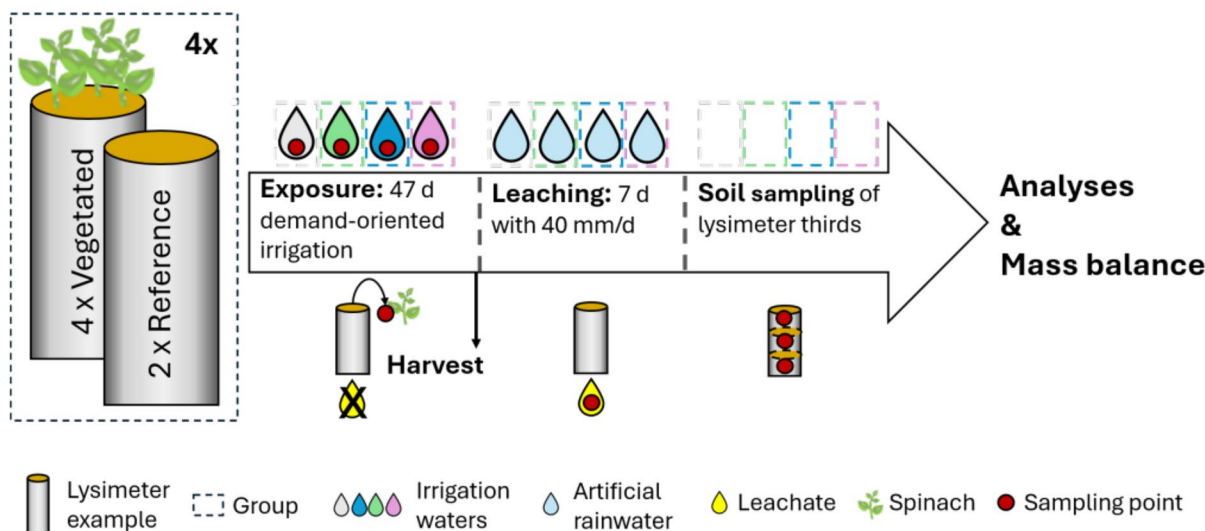


Fig. 1 Schematic of the experimental setup and phases, including exposure, leaching, and soil sampling (retention and extraction). Exposure groups (indicated by colored dashed rectangles) were defined based on the type of irrigation water applied. All exposure groups underwent the same experimen-

tal phases, and all lysimeter replicates within each group were treated identically, except for the spinach harvest conducted in vegetated lysimeters at the end of the exposure phase. The crossed out leachate symbol indicates that no leachates were produced during the exposure phase

More specifically, the investigated OMPs span a physicochemical range, with log D values (calculated at pH 5.3) ranging from -2.71 to 2.77 with a mean of -0.13 , and molecular weights ranging from 150.08 to 613.91 g/mol with a mean of 231.89 g/mol. With exception of carbamazepine (CBZ) and diclofenac (DCF), that have log D values above 2.5 , the OMPs investigated are considered polar referring to their log D values. Among the selected OMPs, only CBZ and primidone (PRI) are neutral compounds, while the rest are either anionic or cationic. For substance-specific details see Table S1 in the Supplementary Information. In this study, emphasis was put on polar OMPs, due to their considerate mobility in aqueous environments and to investigate the relevance for processes occurring in the vadose zone. The well studied OMPs CBZ and DCF are targets of the same analytical method and can be used as reference of less polar OMPs.

For irrigation, ultrapure water was spiked to nominal concentrations of 2 , 10 and 100 $\mu\text{g/L}$ per OMP, resulting in three exposure groups, namely *group*₂, *group*₁₀ and *group*₁₀₀. By using ultrapure water, exclusion of variable wastewater matrix allowed investigation of fate mechanisms of the OMPs under controlled boundary conditions and enhanced reproducibility. The selected exposure levels were chosen to ensure quantifiable concentrations in all sample matrices for reliable mass balance calculations, to assess potential concentration-dependent effects, and to reflect a range of exposure scenarios considering reported occurrence of OMPs in treated wastewater. OMP mass inputs per lysimeter replicate are provided in Table S2 in the Supplementary Information.

A preliminary experiment using the same lysimeter setup with drinking water irrigation resulted in very low mass inputs of only six of the OMPs investigated (Table S2 in the Supplementary Information). This experiment revealed severe difficulties in terms of mass balance calculations at such low OMP mass inputs, while it confirmed negligible background concentrations in soil and spinach samples and was therefore incorporated as a control for unexposed conditions.

2.2 Experiment Procedure

The exposure phase was conducted in a growth chamber (KBW 400, Binder, Germany) with day and night

phase set to 16 and 8 h. Before seeding, 100 g/m² liquid nitrogen-phosphate-potassium-fertilizer ($1:1:1$ g/g/g) were applied, according to the seed supplier's information (Matador, Kiepenkerl, Germany). After seeding, irrigation with respective waters started. Until germination, the temperature was set to 16 and 12 °C and afterwards rose to 25 and 20 °C during day and night phase. Inside the growth chamber, lysimeters were placed on shelves that were equipped with five cool daylight fluorescent tubes for illumination (Lumilux L18W/865, OSRAM, Germany). At first, the laboratory-scale lysimeters were irrigated over 47 d (exposure phase), subsequently leached with artificial rainwater for 7 d (leaching phase) and finally sampled to investigate retention in soil (retention phase). Since the number of lysimeter replicates did not fit into the growth chamber at once, the experiment was conducted in two cycles. In the first cycle, vegetated lysimeters of *group*₂ were irrigated. Owing to external constraints, the irrigation schedule in the first cycle deviated from the experimental protocol and irrigation was interrupted for one week, resulting in an exposure phase of 54 days.

Irrigation water was poured directly onto the soil surface, avoiding contact between leaves and irrigation water. The mean irrigation volume across all lysimeters was 510 ± 76 mL. Demand oriented irrigation volumes were equal to the amount of gravimetrically determined water loss since the respective last irrigation event, resulting in no leachates during the exposure phase. Thereby, the soil water content was kept between 60 to 80% of the soil's field capacity (19.7% v/v) to avoid water stress. To estimate the lysimeters field capacity, soil cores of 100 mL were constructed like the lysimeters used in the experiment, soaked with water, placed on gravel to allow drainage and let to stand for three days under a cover to avoid evaporation. Details on irrigation and evapotranspiration volumes as well as the irrigation schedule per lysimeter are provided in the Supplementary Information spreadsheet.

Starting on the following day, the columns were leached in seven sequential steps with 400 mL of artificial rainwater per lysimeter and day. Artificial rainwater was prepared according to Siemens et al. (2008), representing a rainwater that is typical for the location of the soil's origin. The irrigation water contained 51 $\mu\text{mol/L}$ ammonium nitrate, 9 $\mu\text{mol/L}$ ammonium sulfate, 11 $\mu\text{mol/L}$ calcium chloride, 1.5

$\mu\text{mol/L}$ potassium pyrosulfate, $4 \mu\text{mol/L}$ magnesium sulfate and $12 \mu\text{mol/L}$ sodium sulfate. The pH was adjusted to 5.0 with diluted nitric acid. In total, 2800 mL were applied, equivalent to four times the lysimeter's pore volume. After leaching, the soil cores of the lysimeters were sampled in three equal segments of 5 cm to investigate the distribution of OMPs along soil passage.

2.3 Sampling and Preparation

Leachates were collected in glass beakers for 24 h intermitting the sequential leaching steps. A fraction of 50 mL was sampled from every leachate, filtered ($0.45 \mu\text{m}$ regenerated cellulose, Macherey–Nagel, Germany) and stored below 4°C until analyses. Afterwards, the beakers were washed thoroughly with ultrapure water and reused.

The soil segments were excavated from the columns and instantly homogenized in polypropylene bags. A subsample of each segment was dried at 30°C for subsequent analyses. The aerial spinach parts were cut directly above the soil surface, dried at 30°C and afterwards ground with a ceramic ball mill (CryoMill, Retsch, Germany). The extraction of OMPs from soil and spinach samples was performed using an adapted method (Riemenschneider et al., 2017), with the procedure modified as follows. Half a gram of soil or one gram of spinach was weighed into centrifuge glass tubes. Then, 3 mL of ultrapure water was added and samples left to stand for 60 min. After addition of 5 mL methanol, that was followed by 20 min ultrasonic bath and 30 min shaking (400 rpm), extracts were centrifuged for 15 min at 2800 rpm. The supernatant was decanted into glass bottles. Afterwards, the extraction was repeated with 5 mL of a mixture of methanol and ultrapure water (1:1 v/v). Both supernatants were combined and then filtered ($0.2 \mu\text{m}$ regenerated cellulose, Macherey–Nagel, Germany). The samples were stored at 4°C until analyses. Extractions were performed in duplicates and triplicates for soil and spinach. The extraction efficacy was determined from standards as follows: Two grams of a soil or spinach sample, that was not exposed to the investigated OMPs before, were spiked with 1 mL of an aqueous standard solution, cautiously mixed on a vortex and dried at 30°C . Spinach samples had an extra addition of 3 mL acetone. OMP specific efficacies of standard extraction were first

calculated as fractions of initially spiked masses (for results see Table S3 in the Supplementary Information) and afterwards applied to the results of sample extractions.

2.4 Quantitation of OMPs

Analyses of OMPs in liquid samples were performed with HPLC–MS/MS with instrument settings described in Zeeshan et al. (2023). Soil extracts and leachates were diluted with ultrapure water considering the OMPs mass application, respectively for the low, mid and high exposure groups: soil extracts were measured undiluted or in five- to tenfold dilution; leachate samples were measured undiluted or in fivefold and 50-fold dilution. All plant extracts were measured in tenfold dilution. Quality controls were obtained from control group samples and measured once in all matrices and corresponding dilutions, with accuracies provided in Table S4 in the Supplementary Information. Leachate quality control samples were prepared using samples of the first three and the last leachate step of a control lysimeter after considering changes of dissolved organic carbon and electric conductivity with leaching steps (Fig. S2 in the Supplementary Information). Correction factors were applied accordingly to measurement results of leachates of days one, two, three and four to seven. With largely high accuracies in undiluted quality controls of soil extract samples ($99.8 \pm 14.2\%$), extraction efficacies determined in undiluted samples were applied to all sample extractions.

2.5 Data Processing and Statistics

Measurement results below limits of quantification were excluded from further data processing and mass balance calculations. Measurements of replicate samples were averaged to arithmetic means before further processing. Analyses of extraction samples from control lysimeters were used to determine background concentrations.

Applied mass input (M_{input} , μg) per OMP was determined, based on concentrations in irrigation waters and irrigation volumes per lysimeter. The fraction of M_{input} recovered from leachates (RL , %) was determined from measured OMP concentrations (c_i , $\mu\text{g/L}$) and elution volumes (V_i , L) across seven elution steps i , using:

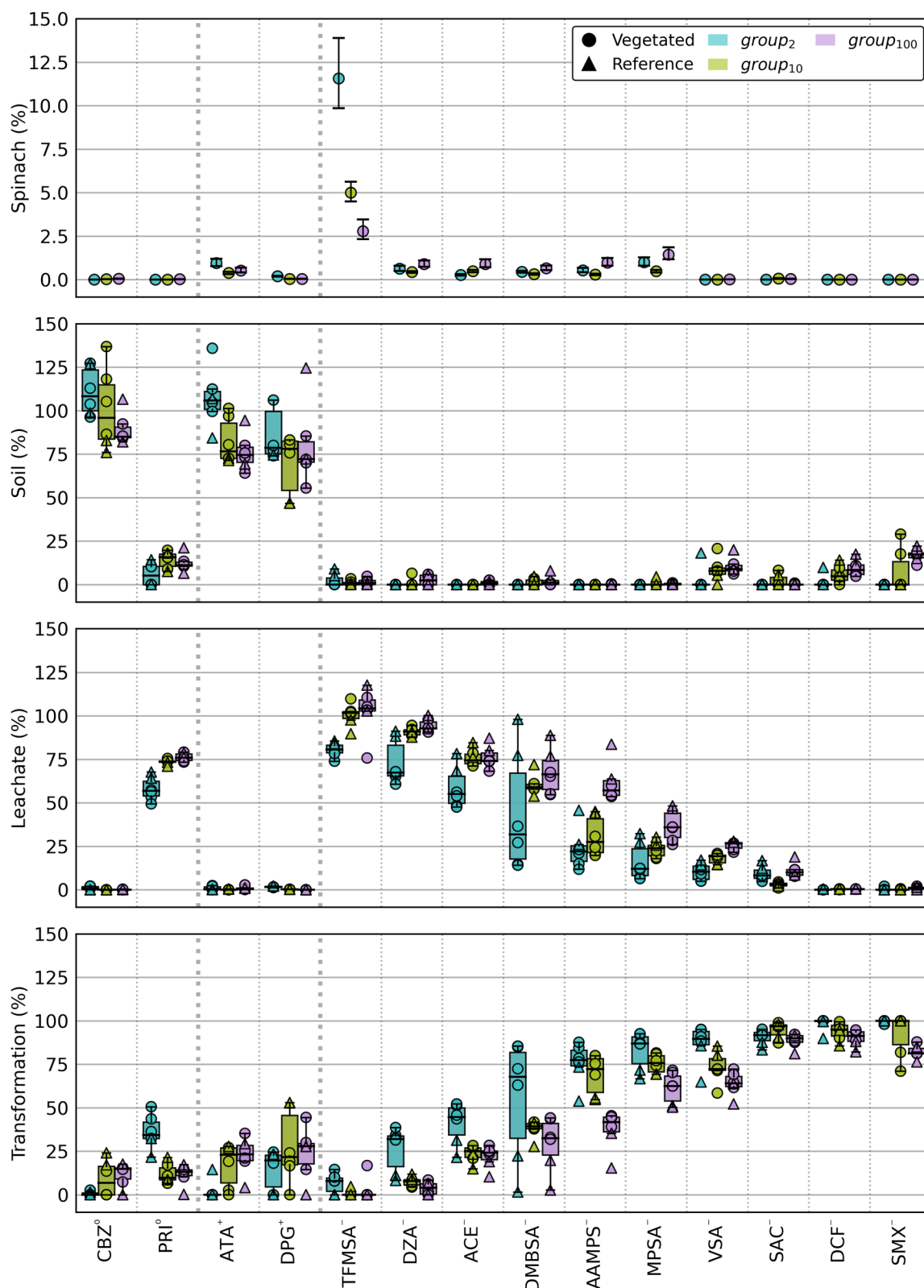


Fig. 2 Mass balance components of OMP recoveries from spinach, soil, leachates, as well as transformation for individual lysimeter replicates. Error bars indicate minima and maxima of plant uptake within each exposure group. Boxplots show median (horizontal line), 25th to 75th percentile (IQR, box), whiskers ($1.5 \times \text{IQR}$) and outliers for exposure groups, including vegetated and reference lysimeters. Negative transformation values were set to zero (original values are provided in the Supplementary Information spreadsheet). The OMPs were grouped according to ionic speciation (neutral (o), cationic (+), and anionic (−)) and subsequently ordered by descending mean total recoveries. The ionic speciation of the OMPs is indicated in the x-axis labels

$$RL = \frac{100\%}{M_{\text{input}}} * \sum_{i=1}^7 c_i * V_i \quad (1)$$

The fraction of M_{input} recovered from soil (RS , %) was calculated as the sum of the products of concentrations (x_j , $\mu\text{g}/\text{kg}$) and bulk density (BD , g/cm^3) and soil volume (SV_j , cm^3), i.e. soil mass, of segment j , using:

$$RS = \frac{100\%}{M_{\text{input}}} * \sum_{j=1}^3 x_j * BD * SV_j \quad (2)$$

Plant uptake per OMP (PU , %) was calculated for respective experiment groups, from the mean concentration (x_s , $\mu\text{g}/\text{g}$) of three replicate samples from the pooled spinach sample, and yielded spinach mass (m_g , g) per experiment group (g), using:

$$PU = \frac{100\%}{M_{\text{input}}} * x_s * m_g \quad (3)$$

Minima of PU were determined for each OMP and exposure group by dividing the minimum concentration spinach of three replicate samples from the pooled spinach sample by the maximum M_{input} among the corresponding lysimeter replicates. Conversely, maxima of PU were determined by dividing the maximum concentration in spinach by the minimum M_{input} . For mass balancing, mean values of PU were used. The unaccounted fraction per OMP is referred to as transformation (T , %) and was estimated based on the following mass balance:

$$T = 100\% - LR - RS - PU \quad (4)$$

Negative values of transformation were set to zero for further processing (for original values per lysimeter see the spreadsheet in the Supplementary Information).

Significant differences among exposure groups were assessed using the Kruskal–Wallis test, followed by Dunn’s post-hoc test for pairwise comparisons when significant differences were detected. The Jonckheere–Terpstra test was applied to evaluate monotonic trends across the groups. Corresponding probabilities were recorded as p_{KW} (Kruskal–Wallis), p_D (Dunn’s) and p_{JT} (Jonckheere–Terpstra). Analyses were performed on leachate and recovery from soil data from vegetated lysimeters ($n=4$ per group), but only for substances where at least two groups exhibited a mean recovery of more than 10%. Mann–Whitney U tests were performed across exposure groups to assess differences between vegetated and reference lysimeters, regarding the fate of OMPs. All data processing was done in *python* (Version 3.8.8). Kruskal–Wallis tests, Dunn’s tests and Mann–Whitney U tests were performed using the *scipy*-package.

3 Results and Discussion

3.1 Mass Balancing and Exposure-Dependent Effects

The obtained mass balances show distinguishable behaviors of either predominantly retained or leached OMPs, while plant uptake minorly contributed to their fate with ratios of mostly less than 3% (Fig. 2). A predominant contribution was attributed when the arithmetic mean of corresponding mass balance terms exceeded 50% across exposure groups. Several substances showed pronounced transformation in combination with leaching.

3.1.1 Predominantly Retained Substances

The three substances adamantan-1-amine (ATA), CBZ and 1,3-diphenylguanidine (DPG) were predominantly retained (Fig. 2), exhibiting mean recoveries from soil of $89 \pm 19\%$, $100 \pm 18\%$, and $85 \pm 31\%$ across the exposure groups. OMP recoveries from soil exceeded 100% in several lysimeter replicates, which is not physically meaningful, thus, reflects uncertainty-related limitations and must be treated cautiously. Nonetheless, such high recoveries are suggested to indicate high persistence and retention potential in soil. Analytical errors are assumed to be minor, based on accuracies of quality control measurements (Table S3 in the Supplementary

Information). It is important to note that the calculation of recovery from soil (Eq. 2) propagates errors, and even small errors potentially result in large deviations from true recoveries, particularly when mass inputs were low. Since no leaching and low plant uptake were observed, transformation remains as second fate pathway besides retention. Among the exposure groups, recoveries of ATA, CBZ and DPG from soil deviate, indicating a challenge of reproducibility. Consequently, transformation must be handled with care as it directly relies on recovery from soil (Eq. 4).

The recoveries of ATA from vegetated lysimeters of exposure *group*₂ were significantly lower compared to those of *group*₁₀₀ ($p_D < 0.02$). Considering CBZ, soil recoveries were close to being significantly lower in *group*₁₀₀ compared to *group*₁₀ ($p_D < 0.07$) and *group*₂ ($p_D < 0.07$). The difference of DPG recoveries from soil between exposure groups was not significant ($p_{KW} > 0.13$). Jonckheere-Terpstra tests did not show significant monotonic trends of recoveries from soil across exposure groups and substances ATA, CBZ and DPG. In summary, no significant impact of exposure levels spanning two orders of magnitude is assumed regarding the recovery of ATA, CBZ and DPG under the experiment conditions.

Observed high recovery of CBZ confirms earlier findings of retention in soil attributed to water reuse (Ben Mordechay et al., 2021; Kodešová et al., 2024; Thalmann et al., 2025a, b) and during soil aquifer treatments (Quintana et al., 2026; Sallwey et al., 2020). Biotransformation might account for observed transformation of CBZ, as it was described by others (Hurtado et al., 2017). Comparable to CBZ, the present study demonstrates retention of ATA and limited transformation in a biotic, unsaturated soil. Recent studies reported sorption of ATA in abiotic batch experiments (Thalmann et al., 2025a, b), but could not decouple biotransformation and sorption in lysimeter studies, using irrigation of reclaimed water (Thalmann et al., 2025a, b). Widespread occurrence of DPG in road soils was reported (Li et al., 2023), as well as in groundwater (Zhang et al., 2025), while data on its occurrence in agricultural soils remain scarce. Complementing observed limited transformation in the present study, persistence in conventional wastewater treatment (Li & Kannan, 2024) and presence in receiving surface waters (Muschket et al., 2024), suggest DPG might pose a risk to soil by irrigation with reclaimed water. Possible mechanisms

explaining the observed retention are discussed in the section on distribution of OMPs across soil depth (3.4).

3.1.2 Predominantly Leached Substances

Trifluoromethanesulfonate (TFMSA), diatrizoate (DZA), acesulfame (ACE), PRI, and dimethylbenzenesulfonate (DMBSA) were predominantly recovered in the leachates following application of artificial rainwater (Fig. 2), with mean recoveries in leachate of $95 \pm 14\%$, $86 \pm 12\%$, $70 \pm 12\%$, $69 \pm 3\%$, and $57 \pm 23\%$, respectively, across experimental groups. Among these substances, only PRI showed measurable retention in soil, with a mean recovery from soil of $11 \pm 6\%$. Consequently, TFMSA, DZA, and PRI exhibited total recoveries exceeding 80%, indicating persistence throughout the experiment. The high variability observed for DMBSA recoveries appears to be driven primarily by the lowest exposure group. In addition to the previously noted deviation of this group from the experimental protocol, reduced analytical accuracy at low exposure (no dilution) may have contributed to the increased variability (Table S4 in the Supplementary Information).

Non-parametric statistical tests including only vegetated lysimeters revealed differences of recoveries in leachates between the exposure groups (Table S5 in the Supplementary Information). Kruskal–Wallis tests indicated significant differences among experiment groups in most cases ($p_{KW} \leq 0.03$). However, Dunn's post-hoc comparisons showed these differences were primarily driven by differences between the lowest and highest exposure levels. In contrast, differences between intermediate and highest exposure levels were mostly not significant. It must be noted that one week of no irrigation, resulting in the prolonged exposure phase (Section 2.2) of vegetated lysimeters of *group*₂, might have led to further impacts than those of exposure level. Nevertheless, Jonckheere–Terpstra tests consistently indicated a significant monotonic trend in leachate recovery across increasing exposure concentrations ($p_{JT} < 0.02$), suggesting a systematic exposure-related effect. The significant difference of the lowest exposure groups suggests a relation to prolonged exposure phase, which might indicate that these substances are susceptible to biotransformation, since no other mass balance

term except transformation substantially varied. Contribution of root uptake remains unknown but is expected to be low, according to limited plant uptake in this study (Section 3.1.4).

In line with the observations, negligible sorption and transformation of DZA was reported for soil batch experiments (Thalmann et al., 2026), supporting earlier reports of DZA findings in groundwater beneath a site irrigated with reclaimed water (Ternes et al., 2007), and tracer-like behavior in lysimeter studies (Thalmann et al., 2025a, b). In contrast to presumably oxic conditions in the conducted experiment resulting from unsaturated soil water contents, DZA was reported to degrade in anoxic environments (Redeker et al., 2014). Possible mechanisms explaining the observed recoveries in leachates are discussed in the section on dynamics of leaching (3.3).

The observed high leaching potential of TFMSA is consistent with a recent lysimeter study (Thalmann et al., 2025a, b). No further reports on the leaching behavior of TFMSA from unsaturated soils were found. Complementing persistence in unsaturated soil, persistence of TFMSA during groundwater recharge (Zeeshan & Ruhl, 2023) and detection of TFMSA in drinking water (Saal et al., 2025) indicate its persistence in aqueous environments.

ACE was completely transformed until soil depth of 10 cm in lysimeter experiments with irrigation of reclaimed water (Thalmann et al., 2025a, b), which is consistent with reports of high transformation in soil incubation experiments (Buerge et al., 2011) and biotransformation of ACE after adaption of the microbiome in batch experiments (Thalmann et al., 2026). The duration of the exposure phase exceeded the reported time for adaption; however, absence of microbial adaption as prerequisite for biotransformation might explain the observed persistence.

Consistent with our observation, PRI was reported to leach from turf grass plots irrigated with reclaimed water (Bondarenko et al., 2012), to be present in groundwater bodies after aquifer recharge (Renau-Pruñonosa et al., 2020) and to be poorly removed during soil aquifer treatment (Sallwey et al., 2020). Moreover, it was already suggested to be included into the European voluntary groundwater Watch List (EU, 2019) as indicator for anthropogenic impact on groundwater bodies. In line with the observed moderate transformation of PRI, Hurtado et al. (2017)

reported the substantial contribution of biotransformation in soils.

3.1.3 Substances Exhibiting Pronounced Transformation

Transformed ratios of 2-acrylamido-2-methylpropane sulfonate (AAMPS), 2-methyl-2-propene-1-sulfonate (MPSA) and valsartanate (VSA) were $61 \pm 20\%$, $73 \pm 12\%$ and $75 \pm 13\%$ across exposure groups. Besides predominant leaching, DMBSA exhibited pronounced transformation, with $41 \pm 24\%$ across the exposure groups. Regarding pathways other than transformation, leaching contributed most to mass balances, in contrast to minor contributions of plant uptake, and recovery from soil observed exclusively for VSA (Fig. 2). Notably, recoveries of VSA and MPSA in leachates were significantly different between $group_{10}$ and $group_{100}$ ($p_D = 0.02$), which were conducted under identical experimental conditions. This strongly suggests concentration-dependent recovery in leachates. Similarly to recoveries in leachate (Section 3.1.2), Kruskal–Wallis tests revealed significant difference of transformation between the exposure groups, while post-hoc Dunn's tests confirmed significant difference only for the $group_2$ compared to the others. Jonckheere–Terpstra tests suggest monotonic exposure-related transformation only for MPSA. To the best of our knowledge, no earlier studies reported on the fate of DMBSA, AAMPS and MPSA in unsaturated soil environments. Reported transformation of VSA in soils (Menacherry et al., 2023) and during soil aquifer treatment (Quintana et al., 2026; Sallwey et al., 2020) are in line with our results.

Throughout the experiment, saccharine (SAC), DCF and sulfamethoxazole (SMX) were transformed to more than 90%, which is consistent with reported high transformation potentials in soil (Buerge et al., 2011; Denora et al., 2024; Facey et al., 2018; Montemurro et al., 2019; Narciso et al., 2024) and during soil aquifer treatment (Quintana et al., 2026; Sallwey et al., 2020).

It should be noted that transformation does not always imply degradation, as transformation products might form and afterwards possibly transform further or retransform. For example, Barbieri et al. (2012) reported transformation of DCF and SMX in aquifer material as well as retransformation of specific

transformation products when environmental conditions became denitrifying. However, nothing similar can be inferred from the present study, as the environmental conditions were not specifically varied and no transformation products of the investigated OMPs were analyzed.

3.1.4 Plant Uptake

Most of the OMPs exhibited plant uptake of less than 3% (ATA, TFMSA, DZA, ACE, DMBSA, AAMPS, MPSA, CBZ, DPG, PRI, VSA), with TFMSA being the only substance showing uptake between 4 and 14% in the highest and lowest exposure groups. DCF and SMX were the only substances with no detection in plant extracts. For greenhouse cultivation of *Eruca sativa* and application of polar OMPs, including all the herein investigated, Seelig et al. (2025) reported higher plant uptake of, e.g. up to 40% for CBZ and even more than 40% for TFMSA. On the contrary, OMPs such as DCF, VSA and SMX were not or only to low extents taken up by plants according to Seelig et al. (2025) and the present study. Comparable to this, high plant uptake of CBZ and no uptake of DCF and SMX was reported in lysimeter studies (Denora et al., 2024). According to spinach yields of approximately 1 g dry weight per lysimeter, limited spinach growth and reduced water uptake during the experiment might have led to discrepant observations. Moreover, the overall evapotranspiration was low, in some cases even observed to be lower in vegetated lysimeters compared to reference lysimeters (Fig. S1 in the Supplementary Information).

3.2 Impacts of Spinach Cultivation on the Fate of OMPs

Mann–Whitney U tests were performed on pooled data across exposure groups to compare vegetated and reference lysimeters. To account for the deviation from experiment protocol of vegetated lysimeters of *group*₂ (Section 2.2), and to assess sensitivity, Mann–Whitney U tests were conducted with and without inclusion of this group. Significant differences were observed for transformation of ACE, DCF, DMBSA, and MPSA in both cases, whereas recoveries in leachate differed significantly only when *group*₂ was included. This reflects the higher variability of recoveries in leachate compared with

transformation. This might be explained by small but systematic differences across recoveries or unquantified losses resulting in significant differences in total recovery and transformation even when leachate recovery alone is not consistently affected. Overall, the results suggest that spinach cultivation primarily influenced substances showing substantial transformation. Enhanced biotransformation might be an explanation, as previous studies reported stimulation of biotransformation of organic micropollutants by plant root exudates (Brunhoferova et al., 2022; Lu et al., 2017).

3.3 Dynamics of Leaching

Based on equivalence of applied artificial rainwater and collected leachate water volume, the soil water content before and after sequential leaching steps was presumably equivalent to the soil's field capacity. This fast equilibration within one day indicates rapid water transport corresponding to the sandy texture. Exceptionally, the leachate volume of the first elution step was less than the applied volume, since the water content during the exposure phase was kept at 60–80% of the field capacity.

The cumulative recoveries of AAMPS, ACE, DZA, DMBSA, MPSA and TFMSA per elution step rapidly reached a maximum (Fig. 3). After the third elution step, amounting to less than 1.5 pore volumes of leachate, more than 80% and partially up to 100% of the total OMP recoveries in leachate were reached. This result suggests high mobility with percolation water. The anionic character of the named substances (Table S1 in the Supplementary Information) might limit interaction with the mainly negatively charged soil matrix or even cause repulsion (Strawn, 2021). Together with observed high recoveries in leachate (Section 3.1.2), mostly unreactive transport is assumed for ACE, DZA and TFMSA under the experiment conditions. Consistently, a tracer-like transport was suggested for DZA and TFMSA and reclaimed water irrigation in long-term lysimeter experiments (Thalmann et al., 2025a, b). On the contrary, substantial transformation of AAMPS, DMBSA, MPSA, VSA, suggest reactive transport. Based on leaching dynamics that are consistent across the exposure groups, it is assumed that neither the level of exposure nor the pathway of transformation affects the leaching dynamics. Detailed investigations of

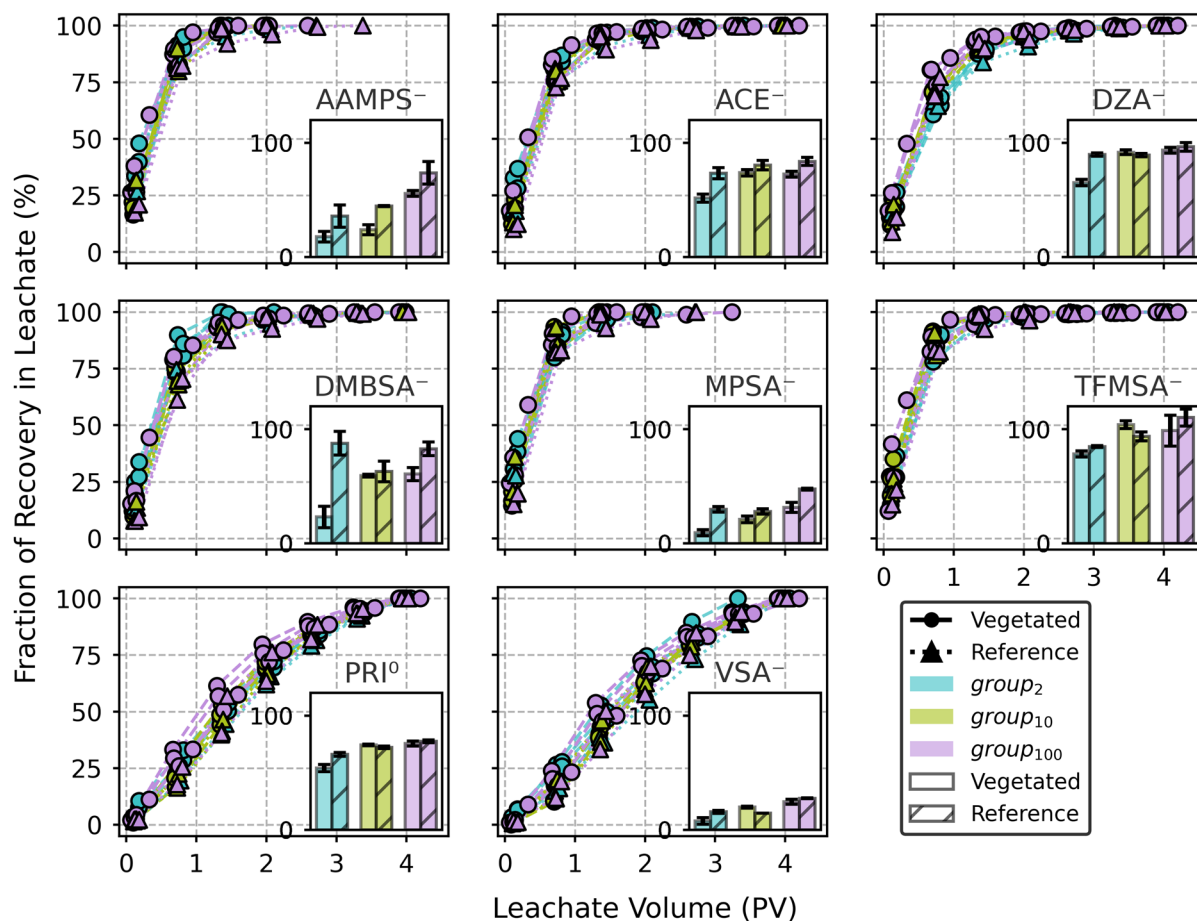


Fig. 3 Fractions of total recoveries in leachates versus eluted pore volumes (PV). Inset bar plots show mean OMP recoveries in leachate for vegetated and reference lysimeters separately with error bars indicating minima and maxima. Expo-

sure group names represent target concentrations of 2, 10 and 100 $\mu\text{g/L}$ in irrigation water. The ionic speciation (neutral (o), cationic (+), and anionic (-)) of the OMPs is indicated in plot titles

transport dynamics in unsaturated soils remain scarce in literature. To the best of our knowledge, leaching dynamics of DMBSA, AAMPS and MPSA from an unsaturated soil were reported for the first time here.

PRI and VSA appeared in the first leachate fraction; however, the cumulated fractions increased gradually and did not reach an apparent maximum until the final leaching step (Fig. 3), suggesting retardation during soil passage. This observation is consistent with the total mass balance, showing low recovery from soil for both substances (Section 3.1.1). PRI is neutral under the experimental conditions (Table S1 in the Supplementary Information), possibly reducing electrostatic repulsion from negatively charged soil surfaces compared with more

mobile substances. In contrast, VSA is anionic. Thus, reversible sorption might be a result of specific interactions with soil organic matter or cation-bridging to negatively charged soil surfaces (Strawn, 2021). However, resolving the mechanistic of retardation was beyond the scope of this study.

The six OMPs exhibiting largely similar and rapid leaching dynamics, namely AAMPS, ACE, DZA, DMBSA, MPSA, and TFMSA, covered a range of log D values between -2.71 and -0.20 . Thus, no effect of log D variation within this range on leaching dynamics can be inferred from the experiment, although minor differences may not have been resolved due to the volumetric resolution of the leachate samples. Slightly retained OMPs showed higher log D values,

with -0.20 for the anionic VSA and 1.12 for the non-charged PRI. In contrast, moderately polar CBZ and the cationic OMPs DPG and ATA did not leach.

These findings are consistent with Valhondo et al. (2023), who reported high retention of cationic OMPs in sand, strong sorption of neutral OMPs to organic carbon, and generally limited retention of anionic OMPs. Accordingly, a positive correlation between the octanol–water partitioning coefficient and sorption was observed for neutral compounds, whereas no clear correlation was found for anionic species. In line with established adsorption mechanisms (Strawn, 2021), the mobility of anionic OMPs is attributed to weak interactions with the predominantly negatively charged soil matrix. However, the slightly greater retention of VSA compared to DMBSA, despite similar log D values and anionic speciation, as well as the largely similar leaching dynamics of VSA and neutral PRI, indicate that log D and charge alone are not always sufficient to predict substance-specific leaching dynamics.

3.4 Distribution of OMPs Across Soil Depth

From the three substances with pronounced recovery from soil, DPG was exceptionally recovered from the top segment. ATA and CBZ were mainly recovered from the top segment as well but additionally recovered from the mid and bottom segments. Considering the substances exhibiting low recovery from soil, DCF, SMX and VSA were only recovered from the top segment and PRI additionally from the underlying segments but only at mid and high exposure levels (Fig. 4). Pronounced sorption in the top segment might be explained by higher solute concentrations compared to the underlying segments due to sequential irrigation events. Thus, initial sorption might have been higher at chemical equilibrium in the top segment compared to the underlying segments.

Despite high polarity indicated by a log D of 0.73 , the cationic state of DPG (Table S1 in the Supplementary Information) might have resulted in strong sorption by electrostatic interaction with negatively charged surfaces of minerals and organic matter (Strawn, 2021). Similarly, the cationic character of ATA suggests ion exchange as a relevant sorption mechanism despite its polar character with a log D of -1.56 . However, in contrast to DPG, the transport of ATA to the mid and bottom segments suggests

weaker sorption, potentially governed by outer-sphere complexation (Strawn, 2021). This is consistent with recent identification of cation exchange as underlying sorption mechanism of ATA onto soil matrix (Thalmann et al., 2025a, b). Since the applied artificial rainwater was low in cations, mobility of the cationic DPG and ATA during irrigation with reclaimed WWTP discharge remains uncertain. Irrigation with WWTP discharge would most likely result in higher cation concentration, increasing competition for sorption sites. Reversible hydrophobic interaction with soil organic matter might have caused the observed retardation of uncharged non-polar CBZ, as reported by others (Malvar et al., 2020; Paz et al., 2016; Zhang et al., 2010). Consistent with our observation, very limited and reversible sorption was reported for DCF, SMX and VSA (Thalmann et al., 2025a, b). Considering the above-discussed retardation and limited transformation of PRI (Section 3.1.2), its detection in the mid and bottom segments might be explained, whereas VSA possibly transformed during retardation.

Overall, weak and only substance specific interaction between anionic OMPs and the soil matrix is suggested, in contrast to the cationic species. Additionally, higher sorption of neutral, non-polar CBZ compared to neutral, polar PRI complements the discussion on leaching of PRI but not of CBZ. The findings support previous reports on retention of polar OMPs exhibiting different ionic speciation in soil aquifer treatment as well as retention of neutral OMPs increasing with log D (Valhondo et al., 2023).

3.5 Plant Uptake Potential

All investigated OMPs except SMX were detected in spinach samples and three groups with differing tendencies towards plant uptake can be distinguished based on clearly separated concentration ranges across exposure groups (Fig. 5). At lowest exposure, the first group, namely TFMSA, was taken up to concentrations that are higher than any other substance exhibited at middle exposure level. The second group includes MPSA, DZA, DMBSA, AAMPS, ACE, and ATA. The group with the lowest apparent uptake tendency comprises CBZ, SAC, DPG, PRI, VSA, and DCF, none of which was quantified at the lowest exposure level, while all were detected at the highest exposure. Within this group, CBZ and SAC

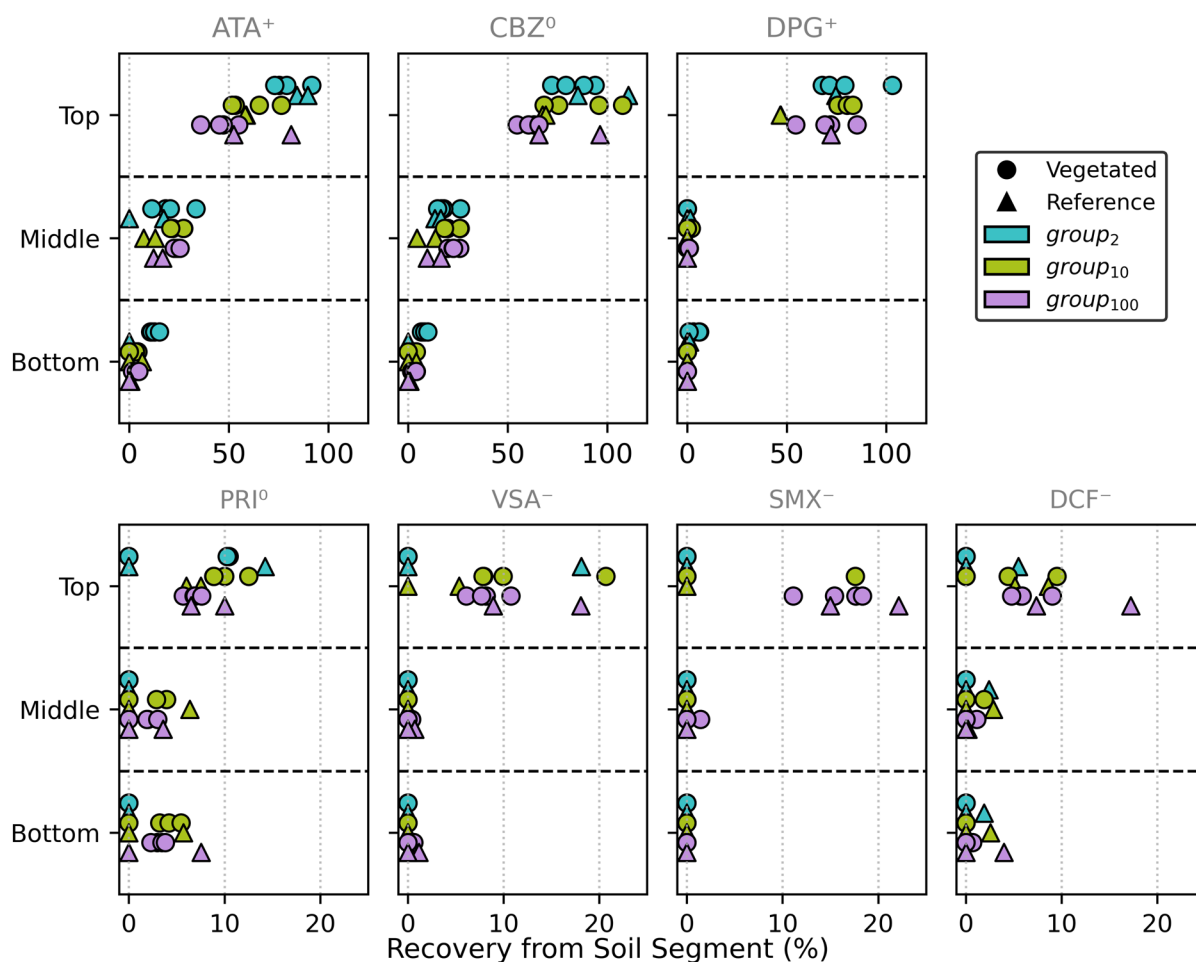


Fig. 4 OMP recoveries from the top, middle and bottom soil segments of each lysimeter replicate. Y-axes are formatted to positions of the soil segments. Exposure group names repre-

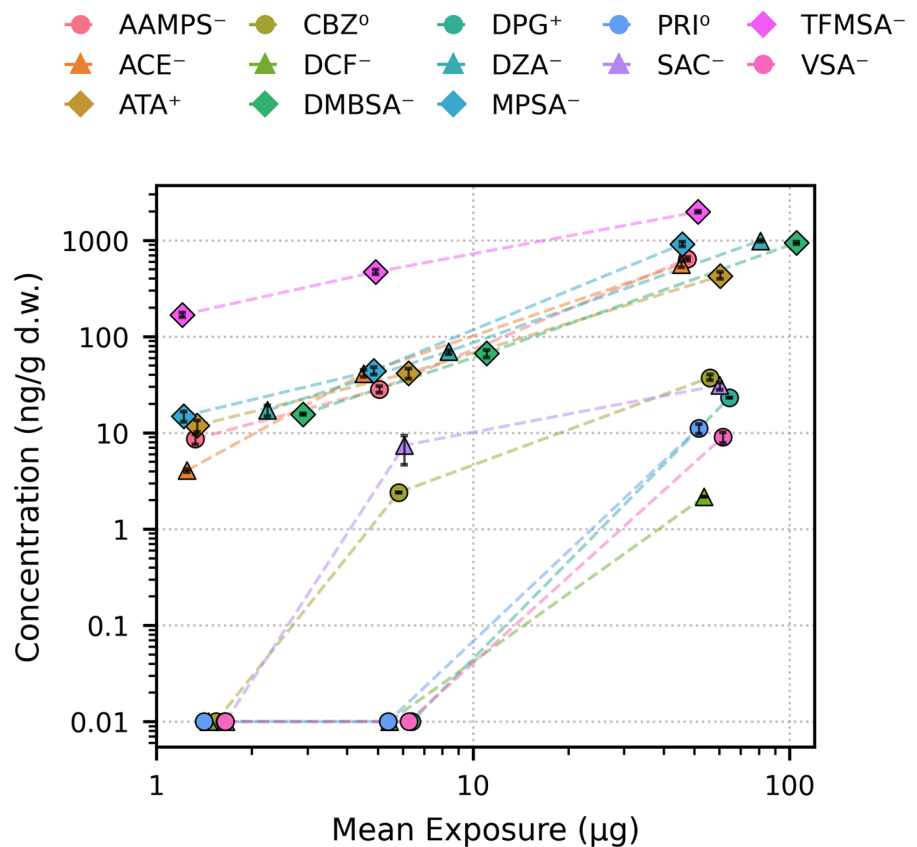
sent target concentrations of 2, 10 and 100 µg/L in irrigation water. The ionic speciation (neutral (o), cationic (+), and anionic (–)) of the OMPs is indicated in the plot titles

were detected at both middle and highest exposure levels, whereas DCF, DPG, PRI, and VSA were only detected in samples of highest exposure. While exposure levels spanned two orders of magnitude, TFMSA concentrations in spinach tissue varied by approximately one order of magnitude, indicating a non-proportional response with increasing exposure. In contrast, such behavior was not observed for the second group of OMPs, which exhibited more proportional increases in tissue concentrations across exposure levels.

According to Wei et al. (2023), diffusive uptake with transpiration water is a major driver of plant uptake of OMPs. Accordingly, the generally high mobility of the investigated substances would be

expected to promote plant uptake, which is broadly supported by the obtained results. Substances that exhibited high recovery in leachate and mobility mostly are in groups of high plant uptake tendencies. However, notable inconsistencies indicate that mobility and pore-water availability alone are insufficient predictors of uptake. For instance, ATA showed comparatively high concentrations in spinach samples despite not being recovered in leachates. Furthermore, while sorption and biotransformation are commonly assumed to reduce availability for plant uptake by decreasing pore-water concentrations (Fu et al., 2019), this relationship was not consistently observed. ATA and CBZ showed similar sorption behavior and persistence (Figs. 3, 4), yet ATA accumulated

Fig. 5 Double logarithmic plot of arithmetic means of mass concentration of OMPs per spinach dry weight (d.w.) versus arithmetic means of exposure within an exposure group. Error bars indicate minima and maxima of triplicate samples. The ionic speciation (neutral (o), cationic (+), and anionic (−)) of the OMPs is indicated in the legend entries



in spinach at substantially higher levels than CBZ (Fig. 5). Similarly, ACE exhibited plant uptake comparable to that of AAMPS and MPSA, although only ACE was persistent under the experimental conditions and all three substances exhibited high mobility (Section 3.3). Seelig et al. (2025) reported markedly different uptake tendencies in *Eruca sativa*, with relative uptake exceeding 40% for TFMSA, more than 30% for CBZ and PRI, approximately 20% for ATA, and less than 5% for substances such as MPSA, VSA, DPG, SMX, and DCF. Discrepancy to the observed results must be interpreted considering contrasting experimental conditions, as Seelig et al. (2025) used reclaimed WWTP discharge for irrigation. With this, a substantially different ionic composition and different mass inputs of OMPs were applied compared to the spiked ultrapure water applied in the present study. Consistent with Seelig et al. (2025), log D was not found to sufficiently predict plant uptake, as for instance apparent for TFMSA, MPSA, AAMPS and ACE in decreasing order of concentrations in spinach (at highest exposure level) and log

D values of -1.23 , -2.21 , -2.71 , and -1.47 , respectively. Overall, the findings support widely discussed assumptions that plant uptake is governed by multiple interacting factors beyond mobility and persistence alone, including substance-specific properties such as ionic speciation and hydrophobicity (Carter et al., 2014; Miller et al., 2016; Santos et al., 2025; Trapp et al., 2023; Wei et al., 2023).

3.6 Implications for Risk Management of Water Reuse

Based on the observed high persistence and sorption, CBZ appears to be a suitable indicator substance for monitoring long-term soil contamination resulting from irrigation with reclaimed water. In contrast, although ATA and DPG exhibited similarly pronounced retention in soil, reported concentrations in WWTP effluents are substantially lower than those of CBZ (Thalmann et al., 2025a, b; Wang et al., 2025), consistent to findings in surface waters receiving WWTP discharge (Muschket et al., 2024).

DZA emerged as a particularly suitable parameter for early detection of groundwater contamination due to its high persistence and mobility. In comparison, reported concentrations of TFMSA in reclaimed water (Muschket et al., 2024; Thalmann et al., 2025a, b) and documented transformation of ACE in soils (Thalmann et al., 2025a, b, 2026), suggest a lower priority for these substances as groundwater monitoring indicators, despite their observed mobility. Notably, DZA was reported to be comparably persistent during advanced treatment processes such as ozonation and activated carbon filtration (Neuwald et al., 2023), further supporting its relevance as a monitoring parameter under water reuse scenarios with and without advanced treatment trains. Employing well-studied substances such as CBZ and DZA as benchmark compounds allows observations from newly investigated substances to be contextualized within water-reuse scenarios, thereby enabling evaluation of their suitability for monitoring and related risks based on irrigation-water concentrations and toxicological descriptors.

Our results confirm earlier reports of a high transformation potential of the vadose zone (Conn et al., 2009). For instance, the observed transformation of VSA and DCF contrast the documented persistence and mobility in riverbank filtration (Zeeshan & Ruhl, 2023) or managed aquifer recharge (Dibyanshu & Scheytt, 2025). Nevertheless, the observed high mobility of several OMPs, in combination with either persistence or moderate transformation, indicates that rapid infiltration into deeper, less bioactive soil layers shortly after exposure may still pose a contamination risk. Consequently, demand-oriented irrigation strategies resulting in longer retention times in biologically active soil layers, together with measures promoting microbial activity, represent effective tools to reduce environmental risks from polar and persistent contaminants.

4 Study Limitations

While sorption, leaching and plant uptake were quantified by mass balancing, the absence of abiotic control lysimeters limits the ability to decouple the role of biotransformation for the unaccounted fraction that is referred to as transformation. As a result, biotransformation can only be assumed, while its quantitative

contribution to overall OMP transformation remains uncertain. The inclusion of abiotic controls in future experiments would allow a more robust separation of physicochemical and biological processes. Furthermore, root uptake that is not followed by translocation within spinach plants, as well as non-extracted residues possibly have contributed to the unaccounted term. However, contributions are suggested to be low regarding overall low plant uptake and high extraction efficacies of predominantly retained substances.

Plant growth and transpiration may not have represented optimal conditions, potentially restricting the evaluation of root uptake, translocation, and bioconcentration. These mechanisms could provide valuable insights into plant-soil-OMP interactions and should be included in future studies. Moreover, the controlled laboratory setup and lysimeter scale inherently simplify natural systems and do not capture field-scale heterogeneity, temporal variability, or dynamic boundary conditions such as fluctuating moisture and temperature regimes. The use of ultrapure water spiked to OMP target concentrations may not precisely cover conditions prevailing in soils irrigated with WWTP discharge in terms of ionic strength of the soil pore water and potential competition (e.g. for adsorption sites) by bulk effluent organic matter. Consequently, direct extrapolation of the results to field conditions should be made cautiously.

The investigation of substances using external calibration standards bears risks of introducing matrix effects into measurement results and subsequent mass balances. However, isotopically labelled internal standards are not always available, especially for newly emerging substances. At the same time, sample matrices, and particularly those of leachates, may vary throughout experiments. Consequently, matrix-matched calibrations require extensive efforts. Furthermore, enrichment of highly polar target analytes from polar aqueous matrices, e.g. by means of solid-phase-extraction, is expected to be poorly efficient. Using elevated exposure levels can therefore help to reduce matrix effects by dilution of samples, while signals remain above limits of quantification.

OMP recoveries from soil exceeded 100% in several lysimeter replicates, which is physically implausible and indicates substantial uncertainty mass balance estimates. Consequently, these values should be interpreted with caution. Nevertheless, the high apparent recoveries mostly suggest that the respective

OMPs were largely retained in the soil, although the magnitude of retention cannot be quantified reliably from these data.

5 Conclusion

This study investigated the fate of fourteen polar OMPs in an unsaturated soil–water–plant system using laboratory-scale lysimeters and spinach cultivation. Using a mass balance approach, the contributions of leaching, sorption, plant uptake, and transformation were quantified, addressing pathways relevant after irrigation with reclaimed water.

Plant uptake was generally low relative to mass inputs; however, three groups of OMPs with varying uptake potentials were distinguished, with TFMSA exhibiting the highest potential. For eight out of fourteen OMPs investigated, near-complete leaching with exchange of one pore volume demonstrated both high mobility and the capacity to bypass transformation during rainfall-driven flush events. For these OMPs, transformation remains the most important process to protect groundwater resources. In contrast, selected cationic polar OMPs, such as ATA and DPG, showed enhanced soil retention, highlighting the influence of charge-dependent interactions on transport behavior. For OMPs exhibiting pronounced transformation, recoveries in leachates increased with exposure, whereas the more persistent OMPs were not affected. In summary, the results underscore the vadose zone as being both a barrier and a reactive compartment while several polar OMPs pose risks to groundwater due to high leaching potentials.

This work contributes to the understanding of the fates of polar OMPs in agricultural irrigation scenarios and provides an adaptable experimental framework to simultaneously assess sorption, leaching, plant uptake, and transformation — pathways that are critical for risk assessment and management in water reuse. Future investigations may incorporate variations of irrigation water, soil moisture regime, and other environmental conditions as well as designs to derive transformation rate constants suitable for predictive modeling. Although field conditions are not fully covered at this laboratory scale, the use of well investigated substances as benchmark process indicators will help to transfer results to environmental applications.

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Authors' Contributions PH: Conceptualization; Methodology; Investigation; Data curation; Formal analysis; Supervision; Writing – original draft; Writing – review & editing. HM: Investigation; Data curation; Formal analysis; Writing – original draft; Writing – review & editing. ASR: Methodology; Funding acquisition; Project administration; Writing – original draft; Writing – review & editing.

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Data Availability Data sets generated during this study are available from the corresponding author upon reasonable request.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interest.

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